

DECISION INTELLIGENCE SYSTEM

Automated Loan Yield Analytics · Incentive Intelligence · Geo Performance Platform

Built on MERN Stack | Real-Time Processing | Sub-60 Second Insights

Yield Engine | Incentive Engine | Geo Performance | What-If Simulator | Auto Insights

The Business Problem

Six critical gaps driving manual errors, delayed decisions, and incentive leakages

PROBLEM	BUSINESS IMPACT
Yield Calculated Manually in Excel	Errors, delays, zero real-time visibility
Incentive Leakage Not Tracked	Company loses efficiency — RMs get wrong payouts
No Geo-Level Performance Visibility	Management cannot identify weak zones fast
MSME Logic Misapplied	RMs push MSME even when HL < 100%, wasting effort
No What-If Simulation Capability	Field teams cannot plan to hit the next slab
No Automated Insight Generation	Decisions based on intuition, not data

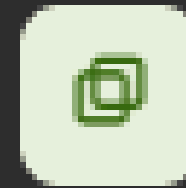


Yield Analytics Engine

Calculates $\text{Yield} = \text{Revenue} \div \text{Disbursal}$ at row level, aggregates to Zone $\rightarrow$ Geo $\rightarrow$ Cluster, and assigns a Yield Score based on basis point gap from target.

bps scoring

Zone / Geo / Cluster

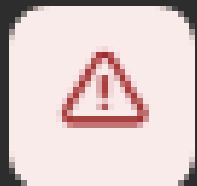


Incentive Slab Classification

Classifies each geo into one of 6 achievement slabs from No Incentive to Maximum, with role-specific rates for RM, BSM, ABSM, and CSM.

6 slabs

RM / BSM / ABSM / CSM

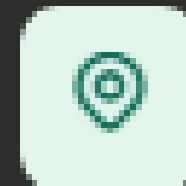


Missed Incentive Intelligence

Calculates exactly how much incentive was left on the table per geo and the precise additional disbursal needed to cross the next slab threshold.

leakage detection

₹ gap per geo

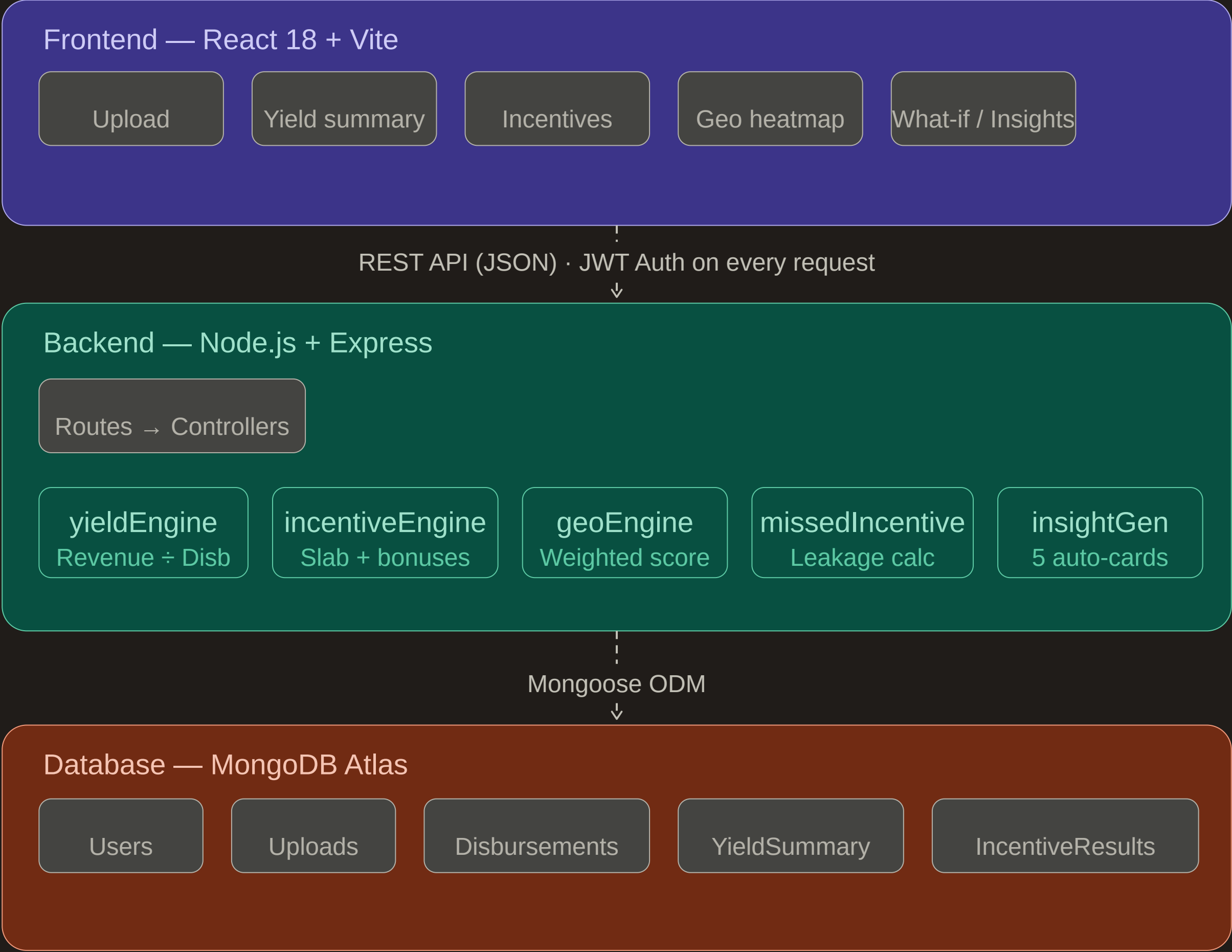


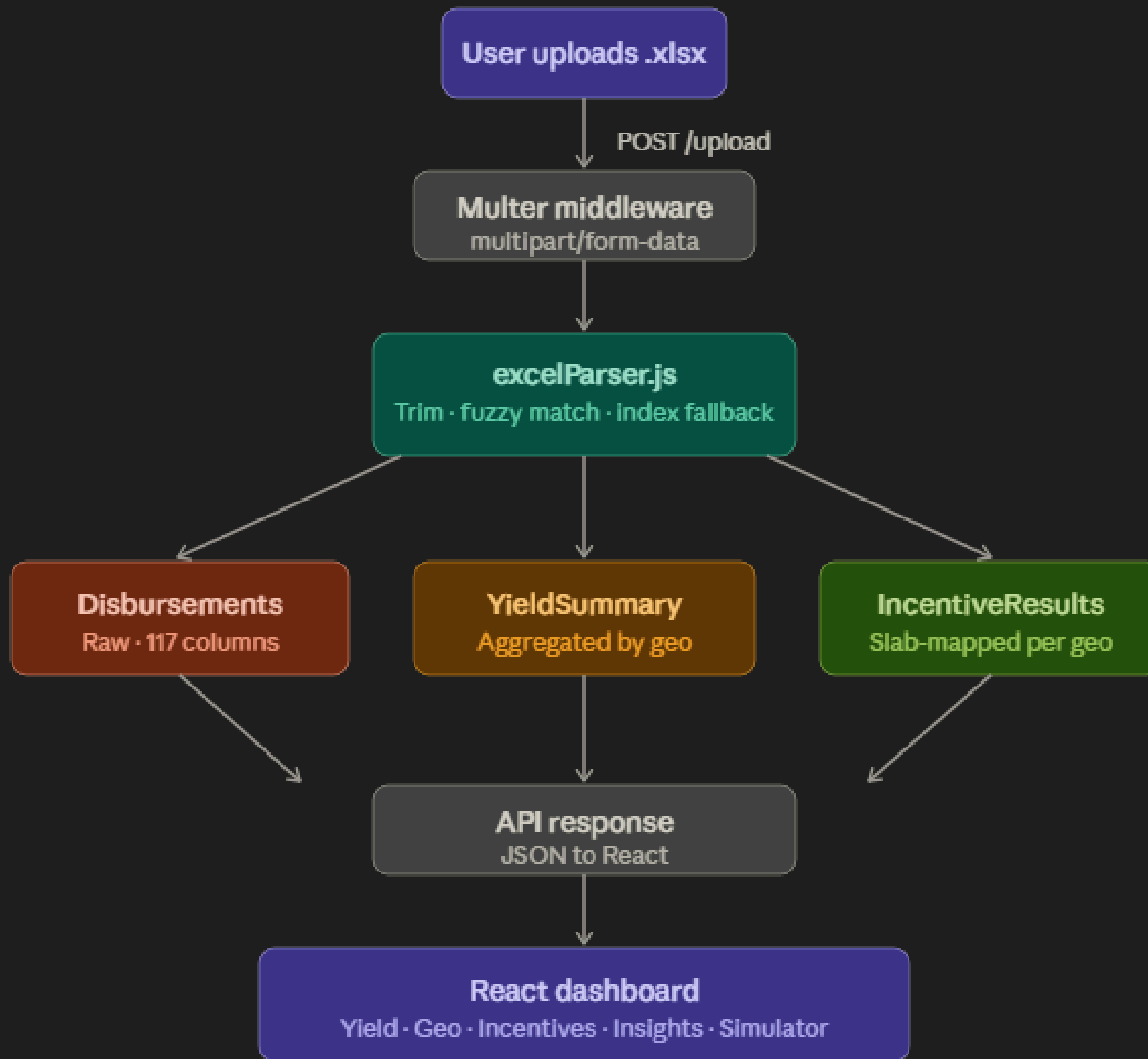
Geo Performance Engine

Scores every geo on a weighted formula — Disbursal 55% + Yield 15% + Conversion 10% — ranks all geos and auto-flags the bottom 5.

weighted scoring

bottom 5 flag





Step 1 — Parse & normalise

Disb Amt (col AO) · Disb ROI (col AQ)

Step 2 — Derived fields

$\text{DisbInCrs} = \text{DisbAmt} \div 1\text{Cr}$

$\text{Revenue} = \text{DisbAmt} \times \text{ROI\%} \cdot \text{WRR} = \text{ROI} \times \text{DisbInCrs}$

Step 3 — Aggregate by geo

$\text{SUM}(\text{Revenue}) \cdot \text{SUM}(\text{DisbInCrs})$ per Zone/Geo

Step 4 — Yield %

$\text{Yield\%} = \text{Revenue} \div \text{Disbursal} \times 100$

Step 5 — Yield score (bps gap vs target)

$\geq \text{target} \rightarrow 125\%$ $<10 \text{ bps} \rightarrow 100\%$ $10-20 \text{ bps} \rightarrow 75\%$

$20-30 \text{ bps} \rightarrow 50\%$ $>30 \text{ bps} \rightarrow 0\%$

Step 6 — Geo score

$\text{Disb} \times 55\% + \text{Yield} \times 15\% + \text{Conversion} \times 10\%$

Step 7 — Incentive slab

$<50\% \rightarrow 0\%$

$50-79\% \rightarrow 0.50\%$

$80-99\% \rightarrow 0.80\%$

$100-124\% \rightarrow 1.60\%$

$125-199\% \rightarrow 2.00\%$

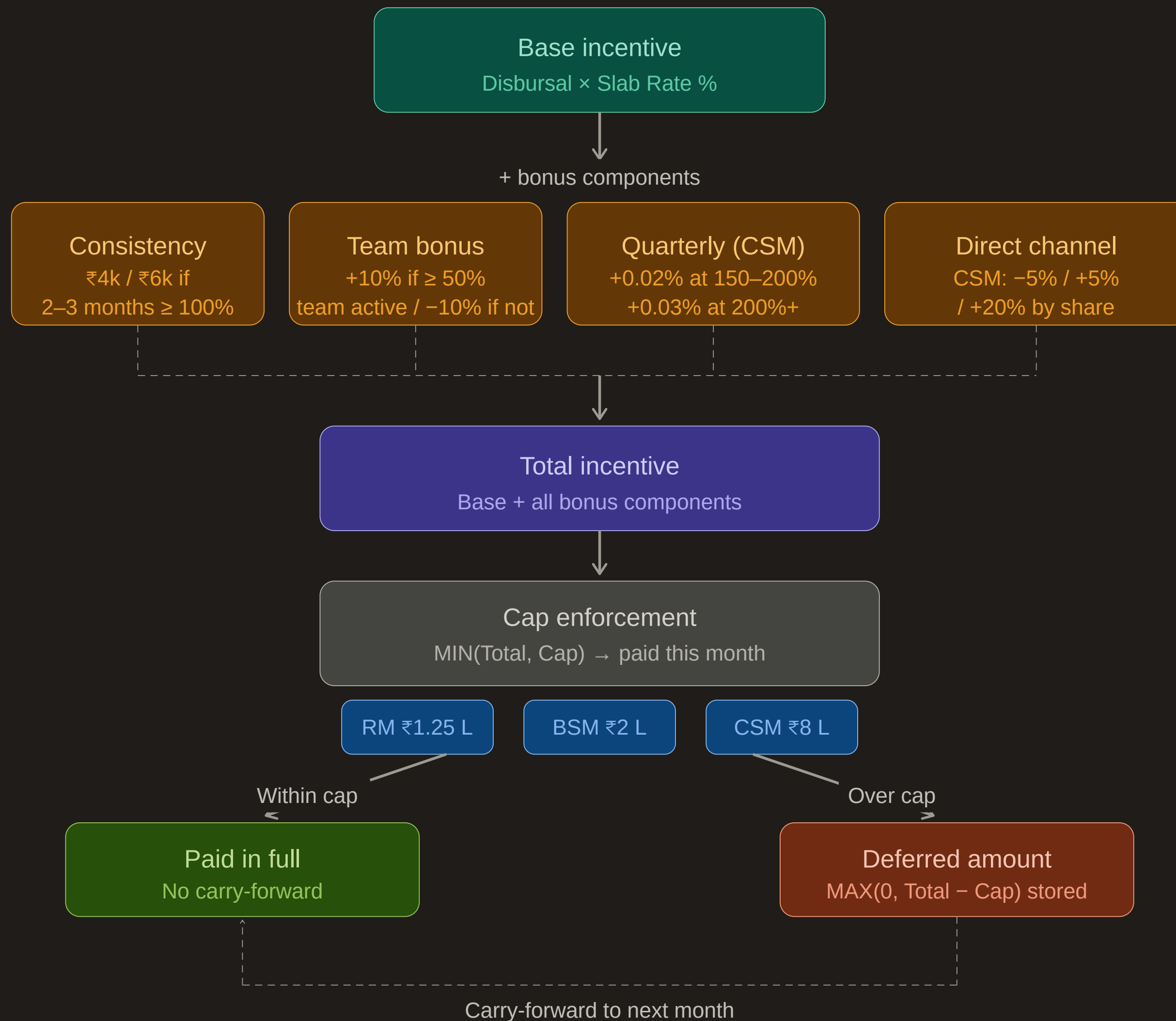
$200\%+ \rightarrow 2.50\%$

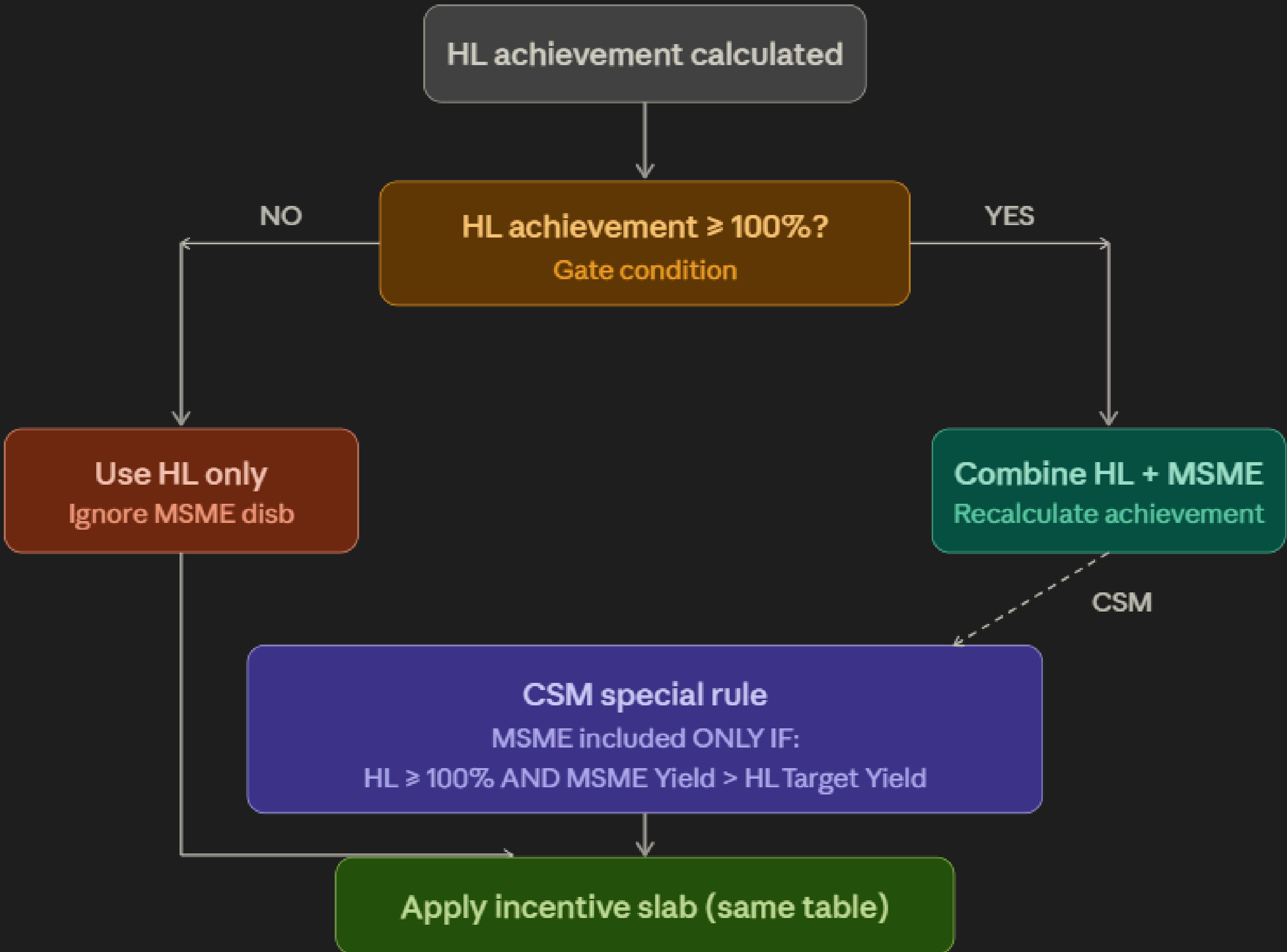
Step 8 — Missed incentive

$(\text{NextRate} - \text{CurrRate}) \times \text{DisbInCrs}$

Step 9 — Auto insights (5 cards)

Yield gap · Disb gap · MSME warning · Leakage · Nudge





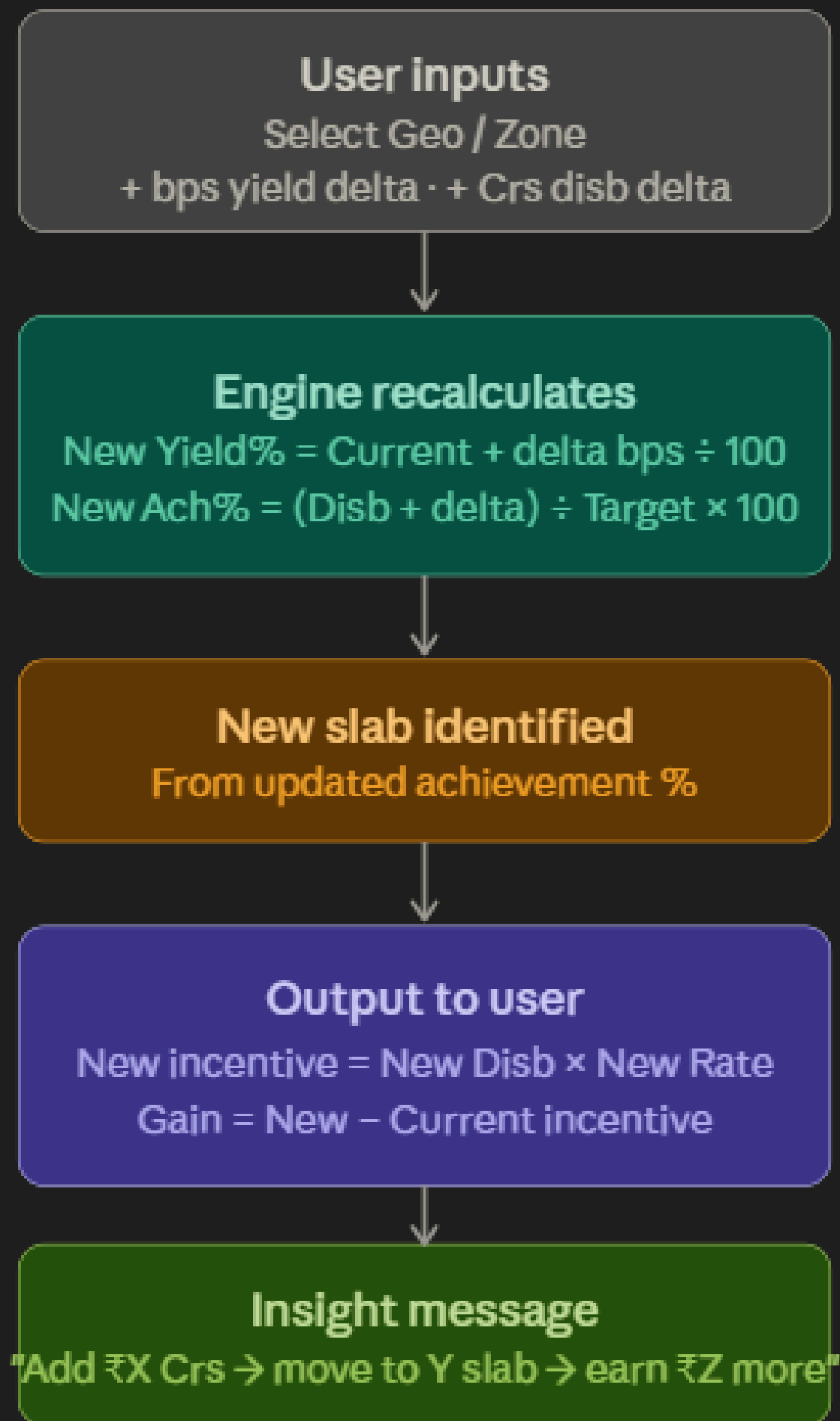
Backend — /server

```
server/  
├─ index.js  
├─ .env  
├─ routes/  
│   ├─ auth.js  
│   ├─ upload.js  
│   ├─ data.js  
│   ├─ yield.js  
│   ├─ incentive.js  
│   ├─ geo.js  
│   ├─ simulate.js  
│   └─ insights.js  
├─ controllers/  
│   (mirrors routes)  
├─ utils/  
│   ├─ excelParser.js  
│   ├─ yieldEngine.js  
│   ├─ incentiveEngine.js  
│   ├─ geoEngine.js  
│   ├─ missedIncentive.js  
│   └─ insightGenerator.js  
├─ models/  
│   ├─ User.js  
│   ├─ Upload.js  
│   ├─ Disbursement.js  
│   ├─ YieldSummary.js  
│   └─ IncentiveResult.js  
└─ middleware/  
    ├─ auth.js  
    └─ roleGuard.js
```

Frontend — /client

```
client/  
├─ index.html  
├─ vite.config.js  
└─ src/  
    ├─ main.jsx  
    ├─ App.jsx  
    ├─ pages/  
    │   ├─ Login.jsx  
    │   ├─ Upload.jsx  
    │   ├─ DataPreview.jsx  
    │   ├─ YieldSummary.jsx  
    │   ├─ Performers.jsx  
    │   ├─ Incentives.jsx  
    │   ├─ GeoHeatmap.jsx  
    │   ├─ Simulator.jsx  
    │   └─ Insights.jsx  
    ├─ components/  
    │   ├─ Sidebar.jsx  
    │   ├─ FilterPanel.jsx  
    │   ├─ InsightCard.jsx  
    │   └─ ProtectedRoute.jsx  
    ├─ hooks/  
    │   ├─ useAuth.js  
    │   └─ useFileId.js  
    └─ api/  
        └─ axios.js
```

What-if simulator



Geo score formula

